Report of CPT205 Assessment 2

0. Basic information

Module Code/Title	CPT205 Computer Graphics
Name	Jiaqi Wang
Student ID	2362441
Degree Programme	Information and Computing Science

1. Brief Description

The 3D Modelling Project shows a futuristic cyberpunk city where players control a hovering car and travel between tall buildings with neon lights. The main car flies just above a textured road, and the player can move forward, backward, left and right, and also go up or down to change height. On both sides, other hovercars move automatically at higher levels. There are also glowing bridges in the sky and tall buildings in the distance, which make the city look deeper and more three-dimensional. At the beginning, a big welcome message appears in the centre of the screen, and later a small message stays in the corner to remind players that they can right-click to open the control menu at any time.

2. Features

2.1 Texture Mapping and Scrolling Effects

In this project, textures such as the road surface and the sky background are loaded manually from BMP files, using a custom **loadBMP()** function. The program reads the pixel information directly into memory and then uses **glTexImage2D()** to upload the data to the GPU. Texture coordinates are mapped with **glTexCoord2f()** so the images align correctly with the ground plane and the sky wall.

The key visual effect is the moving road texture. By updating a texture offset value (**globalTexOffset**) every frame and adding it to the V-coordinate of the UV mapping, the road appears to scroll continuously beneath the hovercar, creating a sense of motion. The road texture uses a circuit-board style image, which helps enhance the cyberpunk and futuristic atmosphere of the scene.

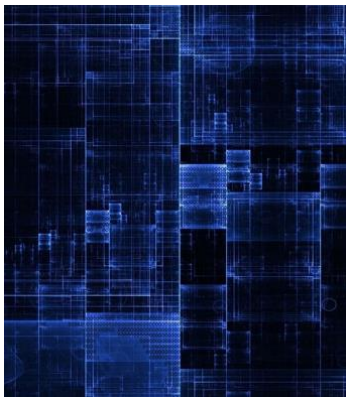


Figure 1. Texture of Road

Instead of traditional skybox, the scene uses a Sky Wall—a large plane textured with a night-sky image—placed far behind the city. To ensure stable rendering of high-resolution textures, bilinear filtering is used.

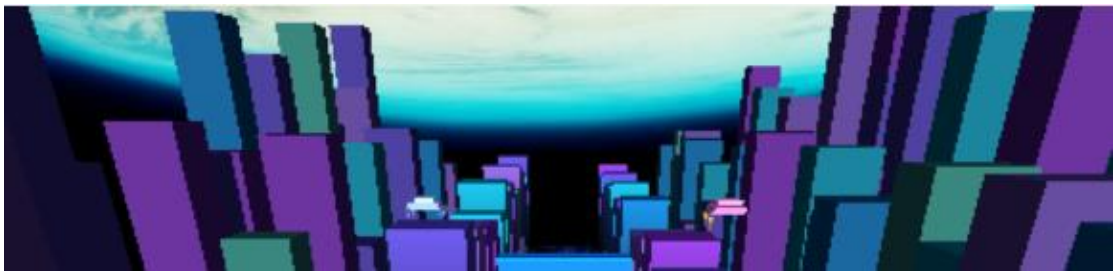


Figure 2 "Sky wall"

2.2 Lighting and Materials

The project uses a single directional light to simulate a distant light source. The ambient, diffuse and specular components are finely tuned to enhance the metallic and realistic of the cyberpunk city. Each building is rendered using a gradient material, blending darker colours at the bottom with brighter tones at the top. This design gives the city a colourful appearance without requiring additional light sources or texure mappings. To further support the cyberpunk atmosphere, I created a custom colour palette containing several cool and saturated shades of blue, teal and purple, and used it to assign random colours to the buildings. These colours, stored in the **RGB customColors[]** array, allow each

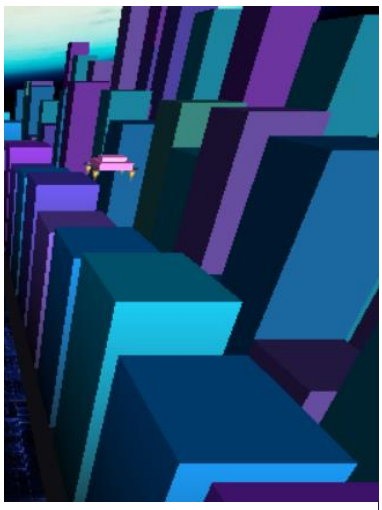


Figure 3 Colourful Building

Figure 4 Hovercars

building block to generate a slightly different tone while still staying within the same futuristic theme. As a result, the city looks more alive and varied.

The hovercar uses several parts with different material styles. Metallic body with specular reflection and cockpit with brighter diffuse colour. Engine flames, rendered with animated cones and semi-transparent blending through **glEnable(GL_BLEND)**.

These lighting and material settings strengthen the atmosphere of a futuristic night city.

2.3 City Generation and Hierarchical Modeling

Both the left and right city blocks are generated procedurally. The seed guarantees reproducible results. Each building receives a random height pattern and a cyberpunk colour from a predefined palette. The hovercar also uses hierarchical modeling, combining multiple solid shapes to form one coherent model. Engine flames animate independently while staying attached to the main body.

2.4 Viewing and Projection

The camera setup uses **gluPerspective()** for a wide 70-degree field of view, creating a sense of speed and vertical depth. The view matrix is updated every frame using **gluLookAt()**, with several dynamic components: As the hovercar rises above the inner-city rooftops, the camera's eye height gradually increases, allowing the player to observe a broader view of the city. Far objects like the sky wall moves smoothly with the camera, giving the illusion of a massive city stretching into the distance. These techniques together create a great flight experience.

2.5 Dynamic Animations

Dynamic behaviour is handled through **glutIdleFunc()**, which updates positions and textures every frame. Key animations include: City

segment recycling (two city blocks switch positions to simulate infinite forward movement), side hovercars that travel at higher altitudes and respawn automatically beyond the visible range, etc.

2.6 Screen Text Rendering

The project uses bitmap fonts (**GLUT_BITMAP_TIMES_ROMAN_24, GLUT_BITMAP_9_BY_15, etc.**) to display on-screen text. At the start of the program, a welcome message and a reminder appear in the centre of the screen with a fade-out effect. After this, a smaller message ("RIGHT-CLICK TO VIEW CONTROLS") remains in the lower-left corner. The smaller message is drawn using 2D orthographic projection so it always stays on top of the scene, regardless of camera movement.

Figure 5 Screen Text

2.7 Menu System and User Interaction

A right-click menu is included to help players understand and use the basic controls. The menu lists all available keyboard actions, so players can check the key bindings at any time and then control the hovercar directly by pressing the corresponding keys. Every menu item is connected to the same movement logic as the keyboard. This means players can also click the options in the menu with the mouse to move forward, turn, ascend, descend, or pause the vehicle without using the keyboard.

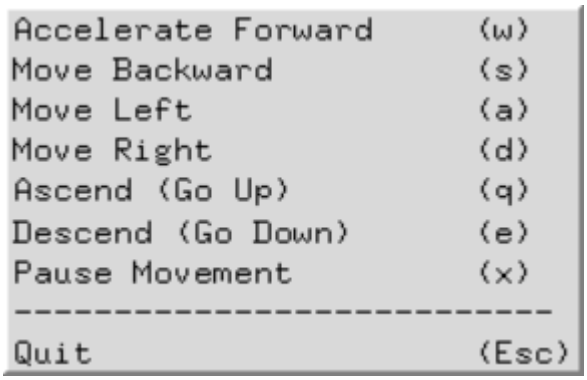


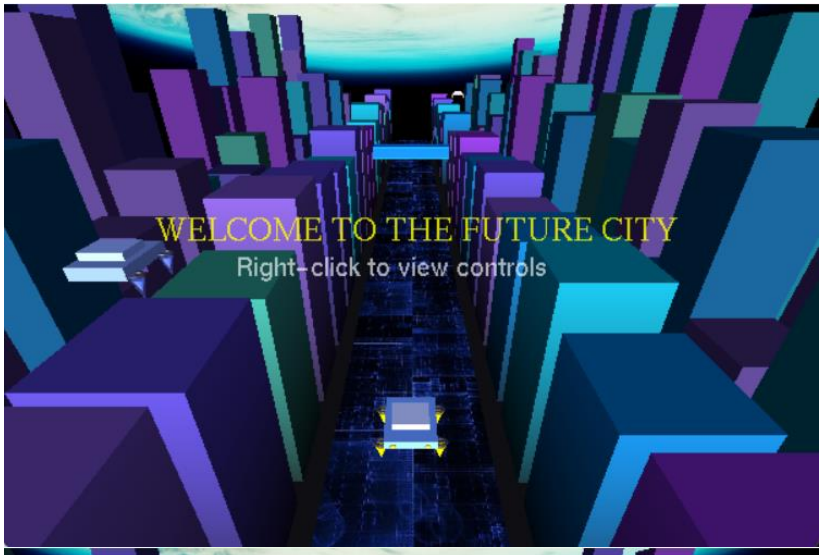
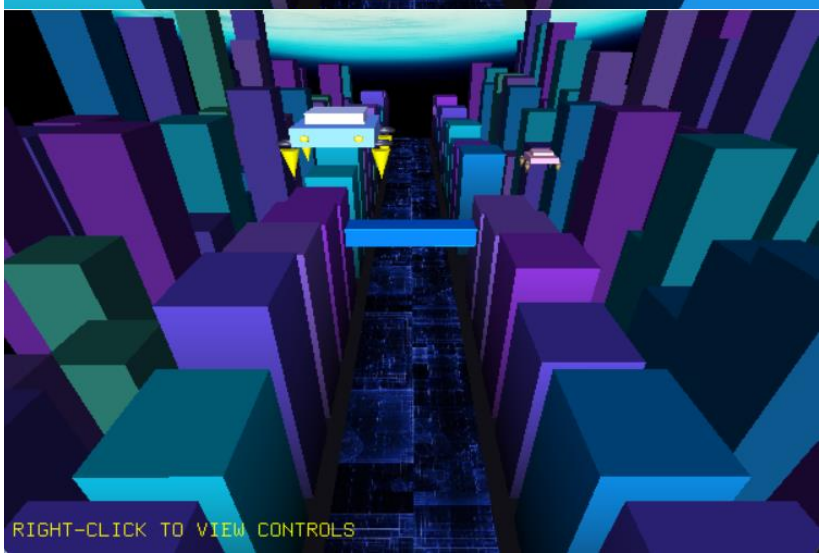
Figure 6 right-click menu

3. Instructions

Open Control Menu: Right-click anywhere to view control options.

Keyboard interaction

Move Forward	W	**Move Backword**	S

Turn Left	a	**Turn right**	d
Ascend	q	**descend**	e
Pause Movement	x	**Quit the game**	Esc

4. Screenshots